

Search for an Optimal Rapid-Charging Pattern for Li-Ion Batteries Using the Taguchi Approach

Yi-Hwa Liu, *Member, IEEE*, and Yi-Feng Luo

Abstract—Lithium-ion batteries are playing important roles as energy storage solutions for portable devices, automotive electronics, and renewable energy systems. In order to maximize the performance of lithium-ion batteries, an advanced rapid-charging pattern is required. In this paper, a Taguchi-based algorithm is presented. Orthogonal arrays are implemented to determine the optimal rapid-charging pattern for multistage constant-current (CC)-charging method. Experimental results show that the obtained rapid-charging pattern is capable of charging lithium-ion batteries to 75% capacity in 40 min. The obtained pattern also provides 60% more cycle life than the conventional CC-constant-voltage method.

Index Terms—Optimal charging pattern, orthogonal array (OA), Taguchi method.

I. INTRODUCTION

ENERGY STORAGE technologies bring an additional degree of flexibility to power systems, providing a means of optimizing the way energy is used by decoupling energy production from utilization. Secondary batteries play a significant part in energy storage solutions for modern technology such as portable devices, automotive electronics, and renewable energy systems. For secondary batteries, three chemistries are used extensively: lead-acid, nickel-metal hydride and lithium-ion (Li-ion). Among these, Li-ion batteries are growing at a rapid rate because it exhibits many advantages such as no memory effect, high working cell voltage, low environmental pollution, low self-discharge rate, and high energy density. The recent trend shows that Li-ion batteries are not only widely used in portable consumer electronics but can also be employed in aerospace vehicle applications and power backup for renewable energy sources [1]–[6].

The performance and longevity of Li-ion batteries depend, to a large extent, on the quality of their chargers. The objectives of high-quality charging include high efficiency, long cycle life, and short charging time [7]–[9]. Battery life is frequently measured in terms of the number of charge/discharge cycles obtained. In typical field use, the cycle count specified by the manufacturer is often difficult to achieve, and batteries fall short of their expected performance. Conventional Li-ion batteries' charging occurs in two steps: The battery is charged

at a constant current (CC) until the battery voltage reaches the predefined upper voltage limit, followed by a constant-voltage (CV) charging until the current reaches a predetermined small value. This method is often called CC–CV-charging method. However, CC–CV is not suitable for rapid charging because CV charging seriously extends the charging time and also reduces the cycle life of the battery. To deal with this problem, many battery-charging techniques are investigated and proposed in the literature [10]–[13]. These methods can achieve better charging efficiencies and shorter charging time. However, these methods are complex and often require digital controllers for implementation. Therefore, these charging algorithms are not suitable to be integrated into commercial battery-charging ICs. Additionally, life cycle issues are often not discussed in these research works. The multistage CC-charging algorithm is proven to have the advantages of prolonged cycle life, enhanced charge/discharge energy efficiency, and reduced charging time [14]. Moreover, this method can easily be incorporated into ICs. However, due to the complex electrochemical characteristics of Li-ion batteries, the optimal current value for each stage is very difficult to obtain.

Searching for the optimal charging pattern can be regarded as a combinatorial optimization problem. Such a problem is difficult to solve by conventional methods such as exhaustive search [15]. Global optimization techniques, such as genetic algorithm (GA), ant colony system (ACS), and particle swarm optimization (PSO) methods, have successfully been employed to solve different combinatorial optimization problems [16]–[20]. In [16], ACS is utilized to improve the design efficiency and control performance of a fuzzy controller. In [17], PSO is proposed to solve multiobjective manufacturing grid resource service composition and optimal-selection problems. In [18], PSO is used to assist power quality disturbance classification. In [19], a novel algorithm that combines standard GA with evolution of a queen bee is presented to obtain the optimal controller for boost converters. In [20], the ACS algorithm has successfully been utilized to search for the optimal rapid-charging pattern. The problem of using global optimization techniques is that the parameters of the algorithm (for example, the ant number, the evaporation rate, and the relative importance of trail intensity for the ACS algorithm) are hard to determine. On the other hand, the Taguchi method has widely been used in experimental designs for problems with multiple parameters. The Taguchi method provides a systematic and efficient method for conducting experiments to determine optimal values [21]–[24]. By using orthogonal arrays (OAs), the Taguchi method searches in the parameter space with a small number of experiments [25]–[28]. In this paper, a global optimization

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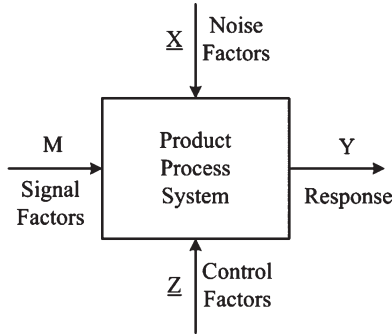


Fig. 1. P-diagram.

technique based on the Taguchi method is introduced for rapid-charging-pattern searching problems. The implementation procedure will be described in detail, and two design examples are presented to demonstrate its validity. Experimental results show that the obtained multistage charging pattern is capable of charging Li-ion batteries to 75% capacity in 40 min. Compared to the conventional CC–CV method, the proposed charging pattern provides 60% more cycles than the regular algorithm, and the charging efficiency is also higher. It is found that the Taguchi method is a good candidate for optimal-charging-pattern searching problems because it is straightforward and easy to implement and can quickly converge to optimal solutions.

II. BASIC CONCEPTS OF TAGUCHI APPROACH

A. Overview of the Taguchi Approach

The Taguchi method is a quality improvement technique proposed by Taguchi. The Taguchi method focuses on improving the fundamental function of the product or process, thus facilitating flexible designs and concurrent engineering. For parameter optimization problems with a given cost function, the Taguchi method is a suitable method for finding the optimal parameter setting and decreasing the number of necessary experiments. The Taguchi method will be described in the following sections.

The Taguchi design consists of four main steps: problem formulation, experiment/data collection, parameter analysis, and prediction/confirmation. In the problem formulation step, a parameter diagram (P-diagram), as shown in Fig. 1, is created first. The P-diagram is used to classify the variables associated with the product into noise, control, signal (input), and response (output) factors. Next, the ideal function and signal/noise (S/N) ratio is defined, and experiments that involve changing the control, noise and signal factors systematically using OAs are designed. S/N is an index of quality that was introduced by Taguchi to determine the optimal combination in a series of experiments. In the experiment/data collection step, experiments may be conducted in hardware or through simulation using the OA developed in the previous step. In this paper, the experiment is carried out in hardware. In the parameter analysis step, the effects of the control factors are calculated, and the results are analyzed to determine the optimum setting of the control factors. In the

TABLE I
 $L_B(C^D)$ OAs

No. of experiments		I	...	...	i	...	...	B
Factors	v_I	v_I^I	...	...	v_I^i	...	...	v_I^B
	$\vdots$	$\vdots$			$\vdots$			$\vdots$
	v_j	v_j^I	...	...	v_j^i	...	...	v_j^B
	$\vdots$	$\vdots$			$\vdots$			$\vdots$
	v_D	v_D^I	...	...	v_D^i	...	...	v_D^B
Experimental results Y		Y_I	...	...	Y_i	...	...	Y_B

prediction/confirmation step, the performance of the system is predicted under the baseline and optimum settings of the parameters. Then, confirmation experiments under these conditions are carried out. If the results of confirmation experiments do not agree with the predictions, the aforementioned steps must be repeated.

B. Experimental Design Using OAs

The Taguchi method was developed on the basis of the OA concept, which offers systematic and efficient characteristics. The first important property of OAs is the fractional factorial characteristic. The second fundamental property of OAs is that all possible combinations of parameters occur equally, which ensures a balanced experiment and permits the cost function of one variable under study to be separable from the cost function of other variables. An additional property of OAs is efficiency. Although balanced, the design of OAs does not require that all combinations be tested. It is demonstrated in statistics that, although the number of experiments is dramatically reduced, the optimum result obtained from the OA is close to that obtained from the full factorial approach.

An OA can be constructed from the knowledge of the number of factors and their levels. In general, the OA can be expressed by $L_B(C^D)$, where B , C , and D represent the number of experimental runs, the number of levels of each factor, and the maximum number of factors that the table can handle, respectively. A typical $L_B(C^D)$ OA is shown in Table I. In general, B is much smaller than the value of C^D , which is the total number of combinations for D factors, with each of the factors having C levels. There are also mixed-level OA $L_B(C^D \times E^F)$, which can deal with D factors with C levels and F factors with E levels.

Once the OA has been determined, experiments can be carried out accordingly, and the measured performance from each experiment can then be used to analyze the relative effect of different factors. In traditional Taguchi's designs, the average value is used to determine the optimal level of the control factors. The average value can be calculated using

$$\overline{Y_{v_j^w}} = \frac{\sum_{i=1}^a \delta(v_j^i, v_j^w) Y_i}{n_{v_j^w}} \quad (1)$$

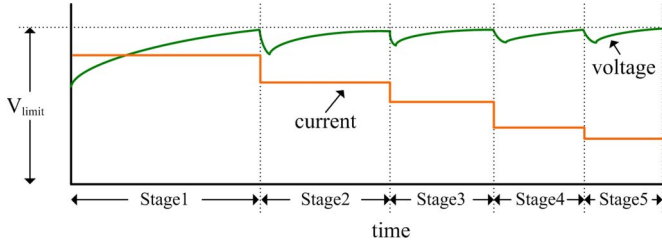


Fig. 2. Multistage CC-charging algorithm.

with

$$n_{v_j^w} = \sum_{i=1}^a \delta(v_j^i, v_j^w)$$

$$\delta(v_j^i, v_j^w) = \begin{cases} 0, & i \neq w \\ 1, & i = w \end{cases}$$

where v_j^i and v_j^w represent the levels of the i th and w th experiments for the v_j th factor, respectively. Moreover, Y_i represents the experimental result for the i th experiment. Depending on the goal of the optimization problem, the optimal level for the v_j th factor can be obtained from (2), where Y_{expected} is the expected value of the system output

$$v_j(\text{optimal}) = v_j^k \begin{cases} \min \{ \overline{Y_{v_j^w}} \}, & \text{smaller the better} \\ \min \{ | \overline{Y_{v_j^w}} - Y_{\text{expected}} | \}, & \text{nominal the best} \\ \max \{ \overline{Y_{v_j^w}} \}, & \text{larger the better.} \end{cases} \quad (2)$$

III. TAGUCHI-BASED RAPID-CHARGING-PATTERN SEARCHING ALGORITHM

A. Problem Formulation

According to the literature, the multistage CC-charging method can achieve the advantages of increased cycling life, shortened charging time, and improved energy transfer efficiency. Therefore, the multistage CC-charging technique is adopted in this paper [14], [29]. The charging time mainly depends on the value of the charging current, and the maximum current for most commercially available Li-ion batteries are lower than 1.5 C (C means C rate and can be defined as a charge or discharge rate that is equal to the capacity of a battery divided by 1 h). Therefore, in this paper, the target charging time is set as 40 min (60 min for 1 C), and the largest current value is set as 1.5 C. Fig. 2 shows the concept of the five-stage CC-charging algorithm used in this paper. From Fig. 2, the total charging period is divided into five stages. In each stage, the charging current is set to a predetermined value. During charging, the voltage of the battery will increase. When the voltage exceeds the preset limit voltage, the stage number will increase, and a new charging-current set value will be applied accordingly. This process will continue until the stage number reaches five.

TABLE II
CHARGING-CURRENT CANDIDATES

Stage	Charging current candidates (unit in C rate)									
1	1.5	1.45	1.4	1.35	1.3	1.25	1.2	1.15	1.1	1.05
2	1.25	1.2	1.15	1.1	1.05	1.0	0.95	0.9	0.85	0.8
3	1.0	0.95	0.9	0.85	0.8	0.75	0.7	0.65	0.6	0.55
4	0.75	0.7	0.65	0.6	0.55	0.5	0.45	0.4	0.35	0.3
5	0.5	0.45	0.4	0.35	0.3	0.25	0.2	0.15	0.1	0.05

A different combination of charging current values will result in different charged-capacity results. Table II shows the charging-current candidates of each stage used in this paper. In Table II, the charging-current candidates in each stage are separated from each other with 0.05 C, and the first five terms of the setting values of each stage are the same as the last five terms of the candidates of the previous stage. From Table II, a full factorial experiment for this searching process would require 100 000 experiments ($100\,000 = 10^5$). However, in order to determine the charged capacity of each combination, the battery should be charged by the charging pattern (a specific current combination) and then be discharged. It should be noted that a constant small current (C/3 or C/10) should be used so as to accurately determine the available capacity in the battery. Consequently, the total test procedure for one charging pattern will last for more than 12 h. Therefore, it is economically inefficient for the manufacturer to test every possible charging pattern to determine the optimal one. To reduce the time consumed in conducting experiments while taking advantage of the performance of the full factorial method, the Taguchi-method-based OA is utilized in this paper.

It should be noted that the current setting of the present stage should not exceed the current setting of the previous stage. Because the voltage-limit value of each stage is the same, if the charging current of the present stage is greater than that of the previous stage, the battery voltage will immediately reach the voltage limit; thus, the duration time of this stage will be very short. In summary, the formulation of this problem can be described as

$$\begin{aligned} & \text{Maximize } \text{ChargedCapacity}(\vec{I}), \quad \vec{I} \in \Omega, \\ & \vec{I} = [I_1 \ I_2 \ I_3 \ I_4 \ I_5] \\ & \text{Subject to } I_i \geq I_j, \quad \text{if } i < j, \quad i, j = 1 \dots 5 \end{aligned} \quad (3)$$

where Ω represents the set of all feasible charging patterns and I_i represents the charging current value of stage i .

B. Taguchi-Based Solution Technique

Fig. 3 shows the flowchart of the proposed Taguchi-based solution techniques. From Fig. 3, the main computation blocks are described in detail in the following.

Step 1) Battery Screening Test

Eighteen brand-new commercially available lithium ion (LiMn_2O_4) batteries are used in this paper. The specifications of these batteries are as follows: nominal voltage of 3.6 V and rated capacity

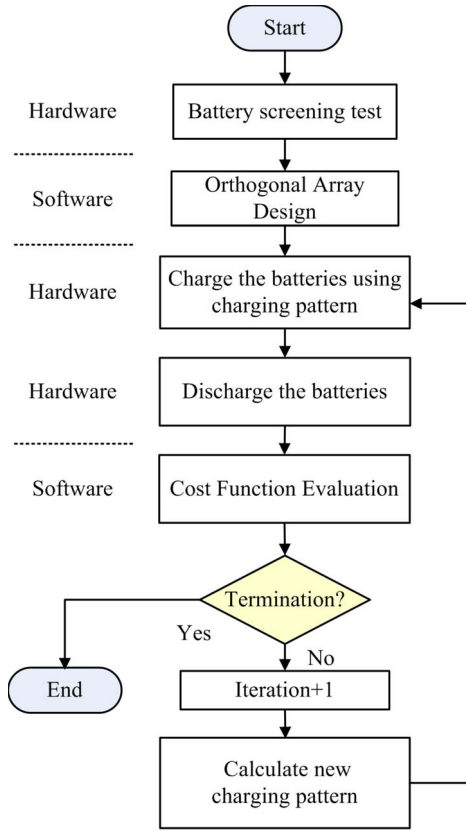


Fig. 3. Flowchart of the proposed method.

of 600 mA · h. All batteries had to pass screening tests with acceptable open-circuit voltage (OCV), impedance, weight, and rated-capacity values.

Step 2) OA Design

An $L_{18}(2^1 \times 3^7)$ OA, as shown in Table III, is selected in this paper. This table is highly recommended by Taguchi, because the interactions between the control factors are quasi-uniformly distributed between the columns of the table. Furthermore, conclusions generated from an L_{18} analysis are nearly independent of the interactions between the control factors. In Table III, numbers 1, 2, and 3 stand for the high, middle, and low levels for each factor, respectively. For example, the setting value of the first L_{18} OA is shown in Table IV.

Step 3) Charge/Discharge Batteries

The charging process was carried out utilizing a WBCS3000 cycle life tester. The charging pattern was read from the file and was programmed into the WBCS3000 battery cycle life tester. After charging the batteries with these charging patterns, the discharging processes are also carried out using the same equipment. Because large discharging current tends to degrade the discharging efficiency, a small discharging current of $C/10$ is utilized to obtain a more accurate total charged capacity. The obtained charged capacity and charging time are then recorded for cost function evaluation.

TABLE III
 $L_{18}(2^1 3^7)$ OA

Number	X	I ₁	I ₂	I ₃	I ₄	I ₅	X	X
1	1	1	1	1	1	1	1	1
2	1	1	2	2	2	2	2	2
3	1	1	3	3	3	3	3	3
4	1	2	1	1	2	2	3	3
5	1	2	2	2	3	3	2	1
6	1	2	3	3	1	1	1	2
7	1	3	1	2	1	3	2	3
8	1	3	2	3	2	1	3	1
9	1	3	3	1	3	2	1	2
10	2	1	1	3	3	2	2	1
11	2	1	2	1	1	3	3	2
12	2	1	3	2	2	1	1	3
13	2	2	1	2	3	1	3	2
14	2	2	2	3	1	2	1	3
15	2	2	3	1	2	3	2	1
16	2	3	1	3	2	3	1	2
17	2	3	2	1	3	1	2	3
18	2	3	3	2	1	2	3	1

TABLE IV
PARAMETER SETTINGS FOR THE FIRST ITERATION

unit(C)	I ₁	I ₂	I ₃	I ₄	I ₅
$\Gamma^+(1)$	1.5	1.25	1	0.75	0.5
$\Gamma^0(2)$	1.4	1.15	0.9	0.65	0.4
$\Gamma^-(3)$	1.3	1.05	0.8	0.55	0.3

Step 4) Cost Function Evaluation

The objective of rapid charging is to charge the batteries as fast as possible without overcharging the batteries. Therefore, the cost function should take both the charged capacity and charging time into account. The cost function used in this paper can be described in

$$\text{Normalization}_i = \alpha \times \left(\frac{C_{i_5 \text{ step}}}{C_{i_CC/CV}} \right) + \beta$$

$$\times \left(\min_C + \left(\frac{\max_T - T_i}{\max_T - \min_T} \right) \times (\max_C - \min_C) \right). \quad (4)$$

In (4), $C_{i_5 \text{ step}}$ and $C_{i_CC/CV}$ represent the charged capacities obtained from the i th battery when using five-step CC- and CC-CV-charging techniques, respectively. α and β stand for the weightings of the charged capacity and charging time, respectively. $\max_T$ and $\min_T$ are the maximum and minimum required charging times of all the batteries in this test run, respectively. $\max_C$ and $\min_C$ are the maximum and minimum charged capacity ratios ($C_{i_5 \text{ step}}/C_{i_CC/CV}$) obtained from the experiments in the first test run, respectively. Furthermore, T_i represents the required charging time for the i th battery. It should be noted that α and β can be set to arbitrary values as long as their summation equals one. In this paper, both

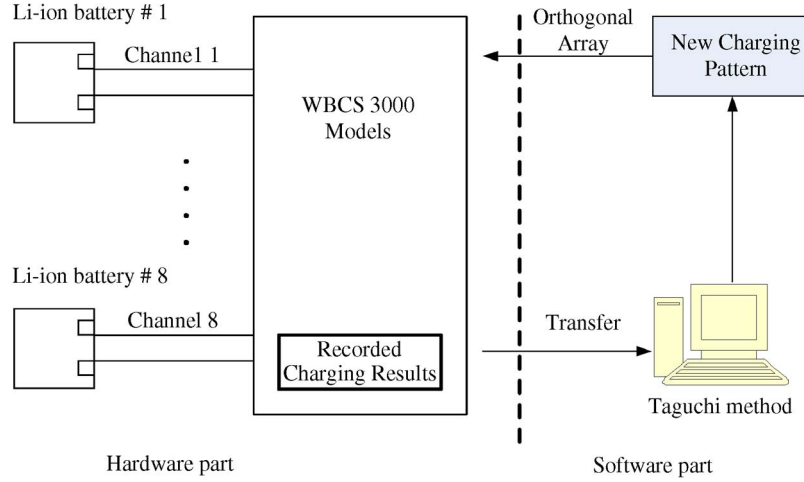


Fig. 4. Experimental system configuration.

α and β are equal to 0.5, i.e., the importance of capacity and time is the same.

Two approaches of how to evaluate the best cost function are investigated in this paper. The averaged value obtained using (1) and the maximum value are utilized to determine the optimal condition.

Step 5) Convergence Determination

After conducting all experiments in the i th iteration, the cost function can be evaluated, and termination criteria are then checked. The level differences of each stage are chosen as the convergence criteria. If the level differences reach the minimum possible value, the process can be terminated; otherwise, the iteration number will increase by one, and a new charging pattern is then calculated.

Step 6) Calculate New Charging Pattern

The new charging pattern is calculated using the following rules, where i represents the stage number ($i = 1-5$), j denotes the iteration number, and the symbols $+$, $-$, and 0 stand for the high, low, and middle levels, respectively.

- If the optimal level is $I_{ij}^0 : I_{i(j+1)}^0 = I_{ij}^0$,
 $I_{i(j+1)}^+ = I_{ij}^0 + 0.05 \text{ C}$, and $I_{i(j+1)}^- = I_{ij}^0 - 0.05 \text{ C}$.
- If the optimal level is $I_{ij}^- : I_{i(j+1)}^0 = I_{ij}^-$,
 $I_{i(j+1)}^+ = I_{ij}^- + 0.1 \text{ C}$, and $I_{i(j+1)}^- = I_{ij}^- - 0.1 \text{ C}$.
- If the optimal level is $I_{ij}^+ : I_{i(j+1)}^0 = I_{ij}^+$ -
 0.05 C , $I_{i(j+1)}^+ = I_{ij}^+$, and $I_{i(j+1)}^- = I_{ij}^+ - 0.1 \text{ C}$.

IV. EXPERIMENTAL RESULTS

Two experiments are carried out in this paper. The only difference of these two experiments is the cost functions used to determine the optimal combination of parameters. The first experiment uses the maximum value of the normalized cost function as the optimal value, while the second experiment utilizes the averaged value of the normalized cost function

as the optimal choice. There are five factors, namely, I1–I5, corresponding to the current setting value of stages 1–5.

Each factor is at three levels. Therefore, an $L_{18}(2^1 \times 3^7)$ OA is utilized. There are 18 experiments required to determine the optimal combination of the levels of these factors, as shown in Table III, and to know the contribution of each to produce the maximum charged capacity and the minimum charging time. To obtain the charged capacity and the charging time, a battery tester is utilized. Fig. 4 shows the configuration of the test system. The charging/discharging equipment used in this paper is WBCS3000 (5 A, 5 V, and eight channels) from WonATech Corporation. Commercially available Li-ion batteries for mobile phones are utilized in these experiments. For Li-ion batteries, OCV and internal impedance are two essential parameters. The nominal OCV should be greater than 3.6 V. As for the internal impedance, a lower impedance will result in lower voltage drop during high-current charging and produce lesser heat, which is therefore favorable. All the batteries used in this paper have passed screening tests, with an OCV that is greater than 4.0 V and an impedance that is lower than 130 m Ω . During the screening test, the initial capacity of each battery is also recorded. This value is important because the charged capacity utilized in the cost function (in percentage) should be evaluated as the capacity obtained from the C/10 discharge process divided by the initial capacity to eliminate the effect of the inherent capacity difference of batteries. The detailed experimental results of the proposed Taguchi method will be provided in the following sections.

A. Taguchi Method Using Maximum Value as Cost Function

The initial parameter settings for the first experiment are listed in Table IV. After implementing the first run of experiments, the maximum value of the cost function is 0.918, and the factor levels that result in the optimal cost function are denoted in Table V using bold characters. The parameter settings for the second run can then be determined from Table V and the rules described in Section III [Step 6)]. Similar processes can be repeated if the convergence condition is not met. Tables VI and VII show the obtained optimal-level setting values after the

TABLE V
OBTAINED OPTIMAL-LEVEL SETTING VALUES AFTER THE FIRST
EXPERIMENT (CASE A, THE OBTAINED COST FUNCTION VALUE: 0.918)

unit(C)	I_1	I_2	I_3	I_4	I_5
$I^+(1)$	1.5	1.25	1	0.75	0.5
$I^0(2)$	1.4	1.15	0.9	0.65	0.4
$I^-(3)$	1.3	1.05	0.8	0.55	0.3

TABLE VI
OBTAINED OPTIMAL-LEVEL SETTING VALUES AFTER THE SECOND
EXPERIMENT (CASE A, THE OBTAINED COST FUNCTION VALUE: 0.921)

unit (C)	I_1	I_2	I_3	I_4	I_5
$I^+(1)$	1.5	1.25	0.9	0.65	0.45
$I^0(2)$	1.45	1.2	0.8	0.55	0.4
$I^-(3)$	1.4	1.15	0.7	0.45	0.35

TABLE VII
OBTAINED OPTIMAL-LEVEL SETTING VALUES AFTER THE THIRD
EXPERIMENT (CASE A, THE OBTAINED COST FUNCTION VALUE: 0.920)

unit (C)	I_1	I_2	I_3	I_4	I_5
$I^+(1)$	1.5	1.25	0.9	0.65	0.45
$I^0(2)$	1.45	1.2	0.85	0.6	0.4
$I^-(3)$	1.4	1.15	0.8	0.55	0.35

second and third experimental runs, respectively. After the third experiment, the range of each level reduces to 0.05 C, which satisfies the convergence condition. Therefore, the process is terminated. The obtained optimal-level setting values for I_1 – I_5 are 1.5, 1.25, 0.9, 0.65, and 0.45 C, respectively. Fig. 5 shows the measured charging current and voltage waveform of this charging pattern. From Fig. 5, the utilized battery tester can accurately perform the charging command.

In order to verify the performance of the obtained charging pattern, charging and cycling experiments were carried out using the CC–CV-charging method and the obtained charging pattern. Fig. 6 shows the measured temperature differences (defined as the surface temperature of the battery under charging minus the ambient temperature) of the CC–CV-charging method and the proposed charging pattern. From Fig. 6, the temperature increase of the proposed charging algorithm is lower than that of the conventional CC–CV method. Because improper charging results in excessive temperature rise in the battery, the charging efficiency of the proposed algorithm can be regarded as higher. Fig. 7 shows the capacity-versus-cycle curve for Li-ion batteries under these two charging algorithms. The curve that is denoted as the proposed algorithm is the cycling test result of the charging pattern obtained in Table VII, and the curve that is denoted as conventional CC–CV is the cycling test result of a CC (1.5-C)–CV (4.2-V) charging pattern. From Fig. 7, the cycling number of the proposed algorithm when the capacity drops to 80% of the original capacity is 185; the cycling number of the CC–CV method is 115. That is, the proposed charging pattern can provide 60% more cycle count than the conventional CC–CV method.

B. Taguchi Method Using Averaged Value as Cost Function

The initial parameter settings for this experiment are the same with that of case A. After implementing the first run of

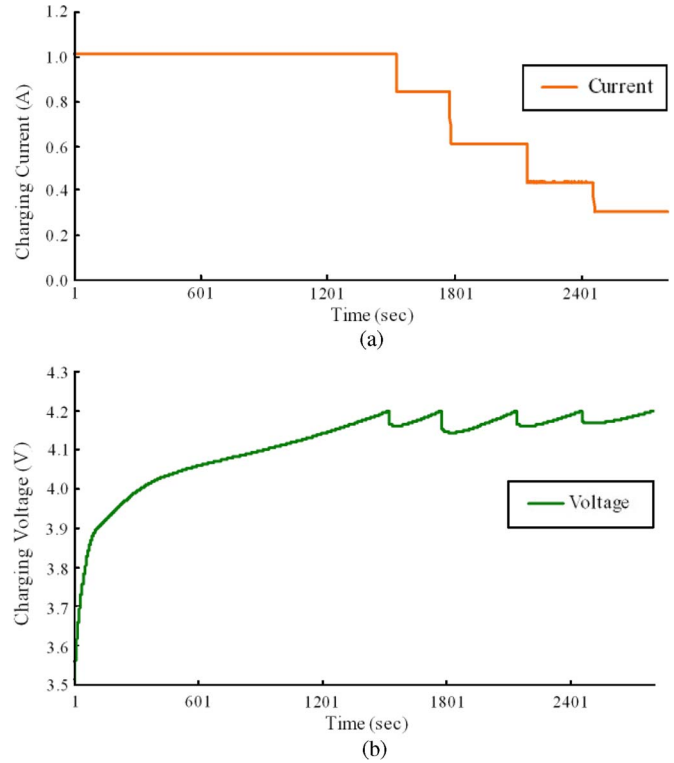


Fig. 5. Measured current and voltage waveform for the obtained optimal charging pattern (1 C = 673 mA). (a) Charging-current profile. (b) Charging-voltage profile.

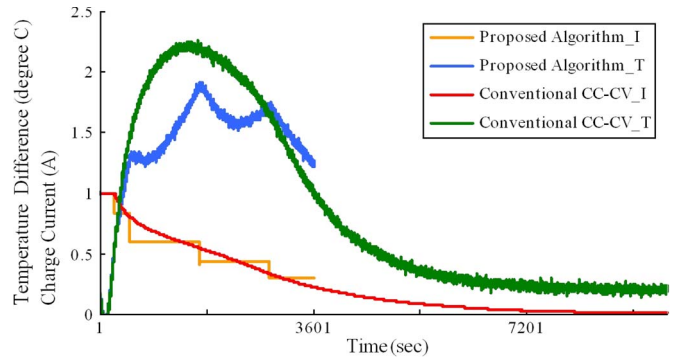


Fig. 6. Temperature difference and charge current of the proposed charging pattern and the conventional CC–CV algorithm.

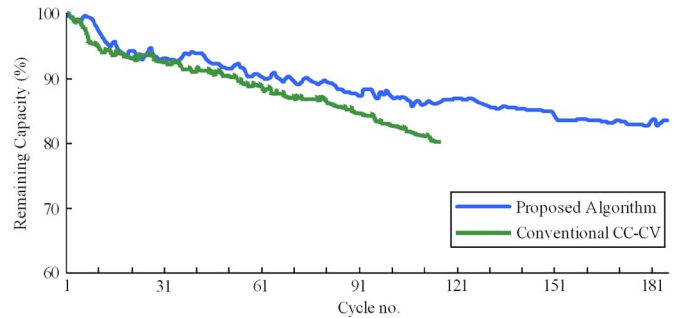


Fig. 7. Cycling test of the proposed and conventional CC–CV algorithms.

experiments, the averaged value of the cost function can be calculated using (1). Fig. 8 shows the main factor effects on cost function defined in (4). It can be seen that the factor-level

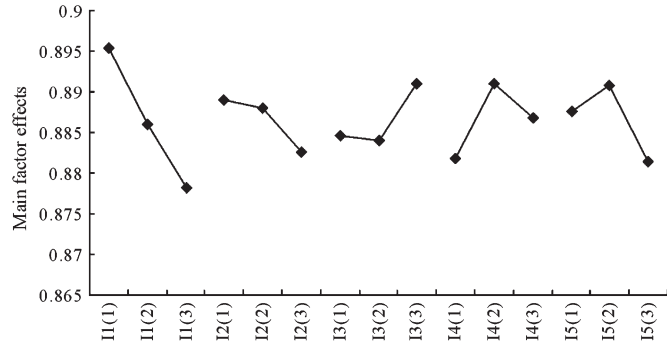


Fig. 8. Main factor effects on maximizing cost function for experiment 1.

TABLE VIII
OBTAINED OPTIMAL-LEVEL SETTING VALUES AFTER THE FIRST
EXPERIMENT (CASE B, THE OBTAINED AVERAGE
COST FUNCTION VALUE: 0.9111)

unit (C)	I ₁	I ₂	I ₃	I ₄	I ₅
I ⁺ (1)	1.5	1.25	1	0.75	0.5
I ⁰ (2)	1.4	1.15	0.9	0.65	0.4
I ⁻ (3)	1.3	1.05	0.8	0.55	0.3

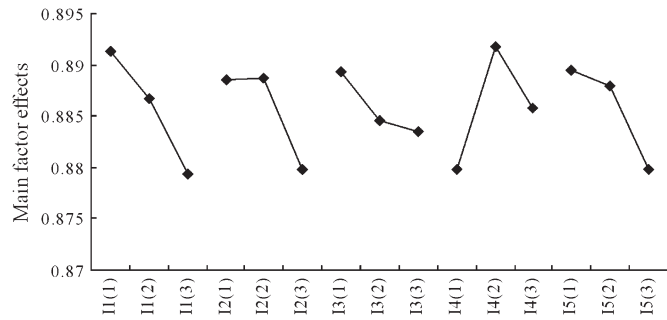


Fig. 9. Main factor effects on maximizing cost function for experiment 2.

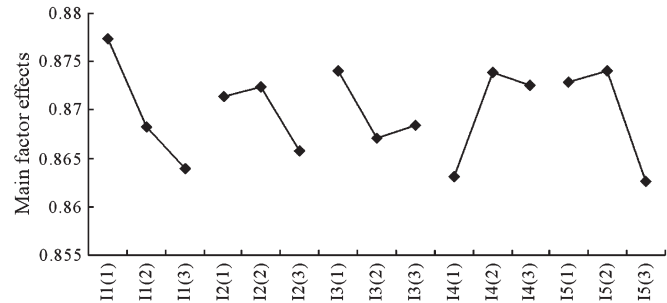


Fig. 10. Main factor effects on maximizing cost function for experiment 3.

combination ($I_1(1), I_2(1), I_3(3), I_4(2), I_5(2)$) contributes to maximize the cost function. These factor levels are denoted in Table VIII using bold characters. The parameter settings for the second run can then be determined from Table VIII and the rules described in Section III [Step 6]. Similar processes can be repeated if the convergence condition is not met. Figs. 9 and 10 show the main factor effects on cost function for the second and third experimental runs, respectively. Tables IX and X show the obtained optimal levels after the second and third experimental runs, respectively. After the third experiment, the range of each level reduces to 0.05 C, which fulfills the convergence condition. Therefore, the process is terminated. The obtained

TABLE IX
OBTAINED OPTIMAL-LEVEL SETTING VALUES AFTER THE SECOND
EXPERIMENT (CASE B, THE OBTAINED AVERAGE
COST FUNCTION VALUE: 0.9076)

unit (C)	I ₁	I ₂	I ₃	I ₄	I ₅
I ⁺ (1)	1.5	1.25	0.9	0.7	0.45
I ⁰ (2)	1.45	1.2	0.8	0.65	0.4
I ⁻ (3)	1.4	1.15	0.7	0.6	0.35

TABLE X
OBTAINED OPTIMAL-LEVEL SETTING VALUE AFTER THE THIRD
EXPERIMENT (CASE B, THE OBTAINED AVERAGE
COST FUNCTION VALUE: 0.8924)

unit (C)	I ₁	I ₂	I ₃	I ₄	I ₅
I ⁺ (1)	1.5	1.25	0.9	0.7	0.45
I ⁰ (2)	1.45	1.2	0.85	0.65	0.4
I ⁻ (3)	1.4	1.15	0.8	0.6	0.35

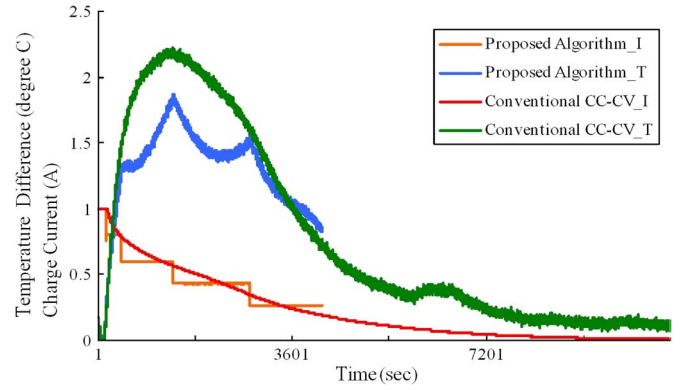


Fig. 11. Temperature difference and charge current test of the obtained and conventional CC-CV algorithms.

optimal-level setting values for I_1 – I_5 are 1.5, 1.2, 0.9, 0.65, and 0.4 C, respectively.

In order to verify the performance of the obtained charging pattern, an experiment was also carried out. Fig. 11 shows the measured temperature differences of the CC-CV method and the proposed charging pattern. From Fig. 11, the temperature increase of the proposed charging algorithm is lower than the conventional CC-CV method. Therefore, the charging efficiency of the proposed algorithm is higher.

To validate this statement, additional experiments were carried out. Table XI shows the experimental data obtained from charging two brand-new batteries (670 mA · h) using the obtained optimal charging pattern and the CC-CV algorithm, respectively. In Table XI, the charging efficiency is defined as the charged capacity divided by the discharged capacity, and the average ΔT is defined as the integration of the temperature differences for the whole charging period (the area under the temperature curve) divided by the charging time required. From Table XI, although the charged capacity of the proposed charging pattern is lower than that of the CC-CV method, the charging time required can greatly be improved. Therefore, the proposed charging pattern can meet the rapid-charging requirement. It can also be observed from Table XI that the charging efficiency of the proposed method is higher than that of the CC-CV method, and the average ΔT is also lower for the proposed charging pattern.

TABLE XI
EXPERIMENTAL DATA OF THE PROPOSED CHARGING PATTERN AND THE CONVENTIONAL CC–CV ALGORITHM

Test Number	CC–CV algorithm					Proposed algorithm				
	Charge Time (sec)	Charged Capacity (mAh)	Discharged Capacity (mAh)	Charging Efficiency (%)	Average ΔT ($^{\circ}C$)	Charge Time (sec)	Charged Capacity (mAh)	Discharged Capacity (mAh)	Charging Efficiency (%)	Average ΔT ($^{\circ}C$)
1	9608	672.10	641.99	95.52	1.931	3614	620.66	597.07	96.20	1.724
2	9581	671.79	641.29	95.46	2.054	3622	620.96	597.48	96.22	1.826
3	9612	672.23	642.05	95.51	2.039	3609	619.88	597.06	96.32	1.760
4	9527	671.24	640.77	95.46	1.921	3575	616.51	593.95	96.34	1.686
5	9573	673.12	642.16	95.40	2.297	3584	617.59	594.99	96.34	1.804
6	9543	671.52	640.09	95.32	1.958	3602	618.41	595.41	96.28	1.743
average	9574	672.00	641.39	95.45	2.033	3601	619.00	595.99	96.28	1.757

V. DISCUSSION AND CONCLUSION

In this paper, an optimization technique based on the Taguchi method has been introduced for rapid-charging applications. The implementation procedure has been described in detail, and two different cost function evaluation methods have been presented. According to the experimental results, both approaches obtain similar results and converge to the optimal solution after three experimental runs by which only 54 experimental tests are needed. The experimental results also show that the obtained rapid-charging pattern is capable of charging Li-ion batteries to 75% capacity in 40 min and provides 60% more cycle life than the conventional CC–CV method.

The proposed technique boasts the following advantages: 1) Compared to other smart rapid-charging methods, the multi-stage CC-charging algorithm can easily be implemented and incorporated into ICs; 2) the Taguchi method is a good candidate for searching the optimal value of the current setting for each stage because it is easy to implement and converges to the desired patterns quickly; and 3) the obtained charging pattern can rapidly charge the battery without causing damage to the battery, therefore prolonging the life of the battery. A further advantage of the Taguchi method is that information can be made available about the interactions between different factors, which is not readily available through other methods. This aspect will be investigated in future work.

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